

VIRTUALIZATION

Module 6

VIRTUALIZATION - DEFINITION

- ❖ The process of **creating a virtual or simulated version** of a physical resource, such as a computer hardware platform, operating system, storage device, or network resource.
 - ❖ A technique to **divide the computer resources logically**. It's achieved by abstracting away the underlying complexity of resource segregation.
 - ❖ It uses software to **create an abstraction layer over computer hardware** that allows the hardware elements of a single computer—processors, memory, storage and more—to be divided into multiple virtual computers, commonly called virtual machines (VMs).
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VIRTUALIZATION - OBJECTIVE

❖ It aims to maximize

❖ resource utilization,

❖ enhance flexibility, and

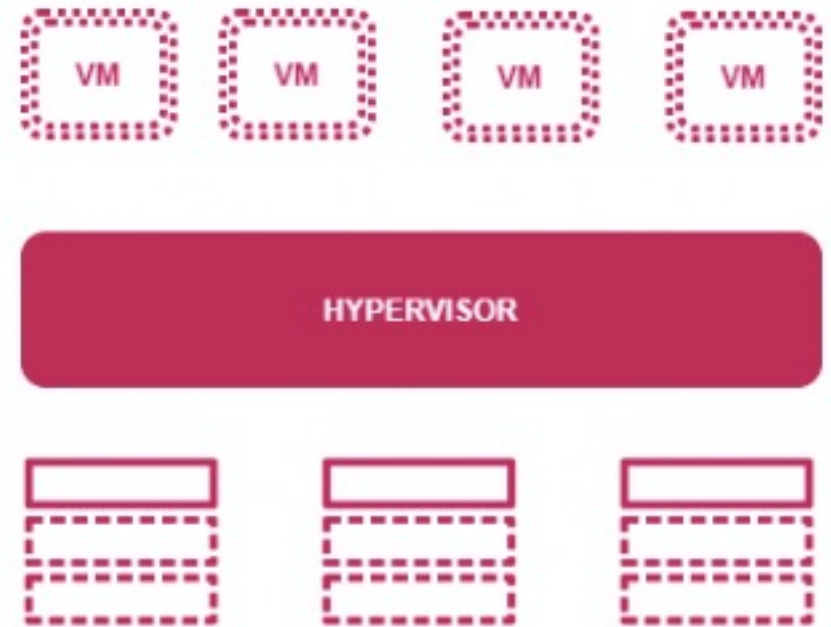
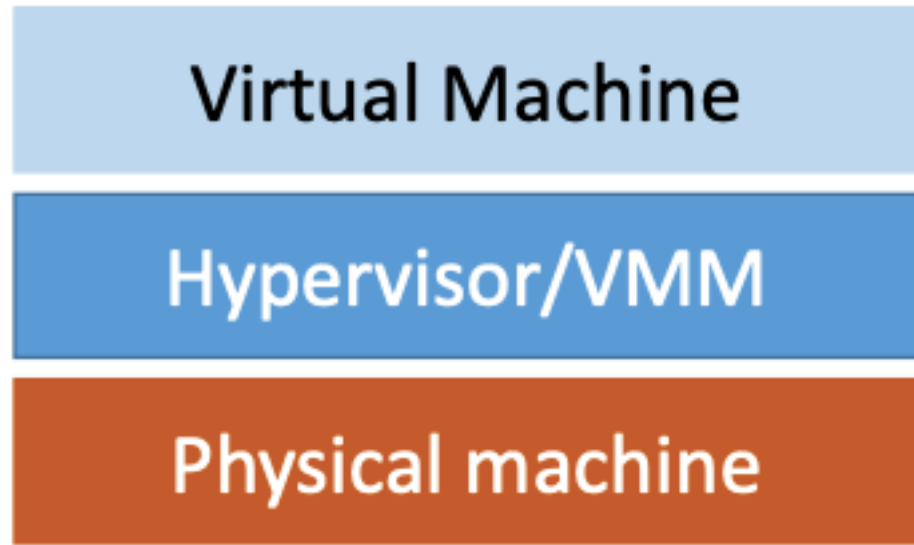
❖ improve scalability.

❖ It enables the abstraction and isolation of resources from the underlying physical infrastructure, providing greater efficiency and agility in deploying and managing software applications and services.

VIRTUALIZATION - VIRTUAL MACHINES

- ❖ It is not a physical machine.
 - ❖ It's a file that replicates the computing environment of a physical machine. (It's like how virtual reality (VR) environments replicate the real world.)
 - ❖ Virtual machine software emulates computer system functions.
 - ❖ A virtual machine (VM) is a software emulation of a physical computer system that behaves like a separate computer with its own operating system and applications.
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VIRTUALIZATION - WORKING



VIRTUALIZATION - WORKING

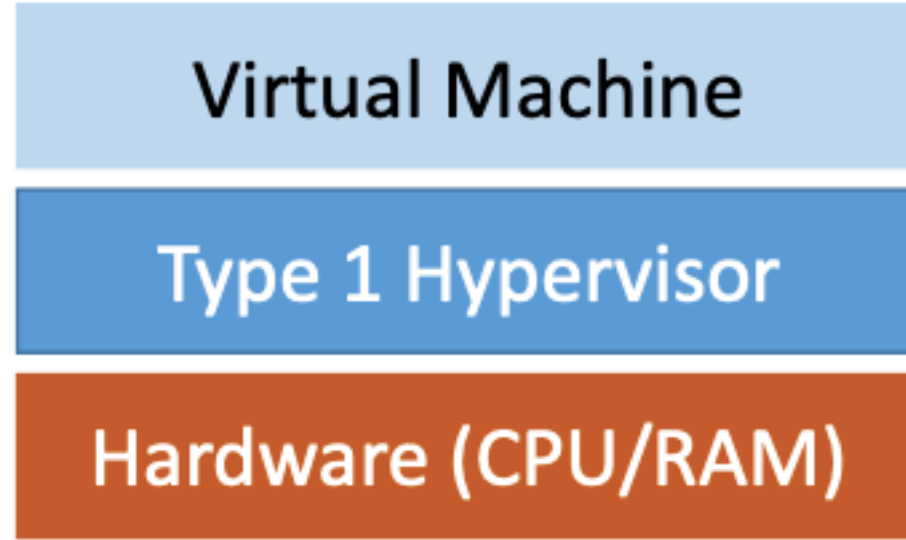
- ❖ Hypervisor software facilitates virtualization.
 - ❖ A hypervisor sits on top of an operating system, but we can also have hypervisors that are installed directly onto the hardware.
 - ❖ Hypervisors take physical resources and divide them up so that virtual environments can use them.
 - ❖ When a user or program issues an instruction to the VM that requires additional resources from the physical environment, the hypervisor relays the request to the physical system and caches the changes.
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VIRTUALIZATION – WORKING - HYPERVISOR

- ❖ A hypervisor is the software layer that coordinates VMs.
 - ❖ It serves as an interface between the VM and the underlying physical hardware, ensuring that each has access to the physical resources it needs to execute.
 - ❖ It also ensures that the VMs don't interfere with each other by impinging on each other's memory space or compute cycles.
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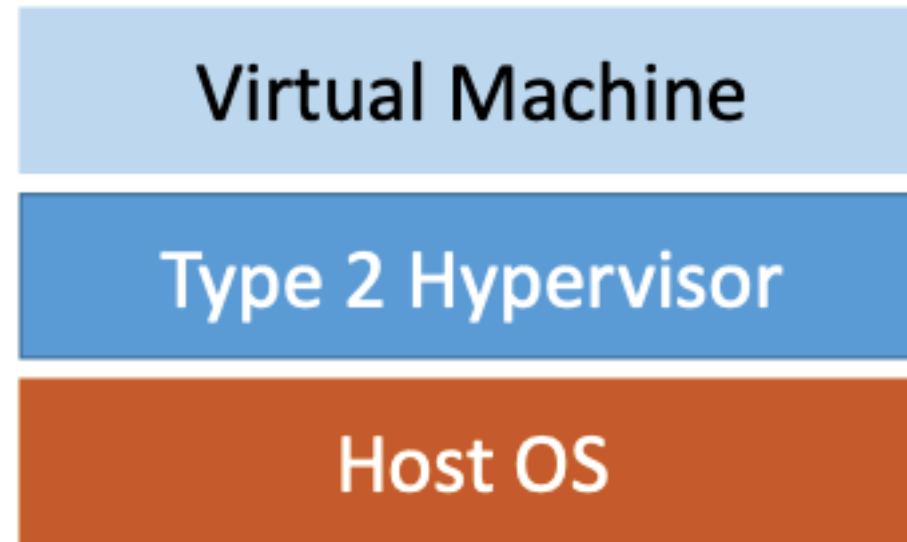
VIRTUALIZATION – WORKING - HYPERVISOR

❖ Type 1 or “bare-metal” hypervisors interact with the underlying physical resources, replacing the traditional operating system altogether. They most commonly appear in virtual server scenarios.



VIRTUALIZATION – WORKING - HYPERVISOR

- ❖ Type 2 hypervisors run as an application on an existing OS. Most used on endpoint devices to run alternative operating systems, they carry a performance overhead because they must use the host OS to access and coordinate the underlying hardware resources.



VIRTUALIZATION – TYPES

❖ Hardware virtualization:

- ❖ This type of virtualization creates multiple virtual machines (VMs) on a single physical server.
 - ❖ Each VM runs its own operating system and applications, as if it were a separate physical machine.
 - ❖ Examples of hardware virtualization technologies include VMware, Hyper-V, and KVM.
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VIRTUALIZATION – TYPES

❖ Server Virtualization:

- ❖ Servers are computers designed to process a high volume of specific tasks so that other computers such as laptops and desktops can do a variety of different jobs.
 - ❖ Virtualizing a server lets it do more of those particular functions and involves partitioning it so that the components can serve multiple purposes.
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VIRTUALIZATION – TYPES

❖ Storage virtualization:

- ❖ Storage virtualization abstracts physical storage resources and presents them as a unified virtual storage system.
 - ❖ It enables centralized management and allocation of storage space from multiple storage devices or arrays.
 - ❖ Storage area networks (SANs) and network-attached storage (NAS) are commonly used in storage virtualization.
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VIRTUALIZATION – TYPES

❖ Network virtualization:

- ❖ Network virtualization decouples network resources from the underlying hardware infrastructure, allowing multiple virtual networks to coexist on the same physical network.
 - ❖ It enables the creation of logical networks, virtual switches, and virtual routers, providing enhanced network scalability, isolation, and management.
 - ❖ Technologies like VMware NSX and Cisco ACI are used for network virtualization.
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VIRTUALIZATION – TYPES

❖ Operating system virtualization:

- ❖ Also known as containerization, this form of virtualization allows multiple isolated user-space instances, known as containers, to run on a single host operating system.
 - ❖ Containers share the same kernel but have their own individualized file systems and process spaces.
 - ❖ Docker and Kubernetes are popular technologies for operating system virtualization.
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VIRTUALIZATION – CONTAINERS

- ❖ They share an underlying OS kernel, only running the application and the things it depends on, like software libraries and environment variables.
 - ❖ This makes containers smaller and faster to deploy.
 - ❖ Containers share the host hardware's OS instead of creating a new version.
 - ❖ Containers avoid creating a separate guest OS, they can hold just the application and its libraries.
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VIRTUALIZATION – CONTAINERS VS VIRTUALIZATION

Virtualization	Containers
Each VM has its own dedicated resources, including CPU, memory, storage, and network. This provides strong isolation between VMs	They share the host operating system kernel and run as isolated user-space instances. They provide lightweight and fast resource isolation
Emulation layer introduces some overhead, which can impact performance compared to running applications on bare metal.	They being lightweight and sharing the host kernel, have minimal overhead and generally offer better performance and faster startup times .
Virtual machines can run different operating systems on the same physical host, allowing for flexibility in terms of OS choices.	They share the same host operating system kernel. This means that all containers running on a host must use the same operating system as the host.
They are more resource-intensive and time-consuming compared to containers	Containers package applications and their dependencies into portable and lightweight images, which can be easily deployed and managed

VIRTUALIZATION – CONTAINERS VS VIRTUALIZATION

Virtualization	Containers
Less Density than containers	Due to their lightweight nature and efficient resource utilization, containers generally have higher density than virtual machines. Multiple containers can run on a single physical server without incurring the overhead of separate operating systems.
These are not highly scalable as containers	Containers are also highly scalable, allowing for dynamic allocation and scaling of resources based on demand
Virtual machines are well-suited for scenarios where strong isolation and running different operating systems are required, such as hosting legacy applications or providing complete virtualized environments.	Containers are ideal for microservices architectures, application development and testing, cloud-native applications, and scenarios where lightweight and fast deployment is crucial.

VIRTUALIZATION – COST OF VIRTUALIZATION

❖ **Performance overhead:**

- ❖ Virtualization introduces a layer of abstraction between the hardware and the virtualized environment, which can lead to a slight performance overhead compared to running applications on bare metal.
 - ❖ While this overhead has been significantly reduced in modern virtualization technologies, certain performance-sensitive workloads may experience a noticeable impact.
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VIRTUALIZATION – COST OF VIRTUALIZATION

❖ **Complexity and management overhead:**

- ❖ Virtualized environments can introduce complexity, particularly as the number of virtual machines or containers increases.
 - ❖ Managing and monitoring a large-scale virtualized infrastructure requires additional effort, specialized skills, and management tools.
 - ❖ Complexity can also arise when dealing with interoperability and compatibility issues between different virtualization technologies or when integrating with existing systems.
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VIRTUALIZATION – COST OF VIRTUALIZATION

❖ **Hardware dependencies:**

- ❖ Virtualization heavily relies on the underlying hardware, particularly the server or host machine.
 - ❖ Compatibility issues or limitations in hardware resources can impact the virtualization capabilities or performance.
 - ❖ It is crucial to ensure that the hardware is properly selected, configured, and maintained to support the virtualized environment.
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VIRTUALIZATION – COST OF VIRTUALIZATION

❖ **Single point of failure:**

- ❖ While virtualization technologies offer features like high availability and fault tolerance, there is still a risk of a single point of failure.
 - ❖ If the physical host machine or the virtualization infrastructure experiences a failure, multiple virtual machines or containers running on it may be affected simultaneously.
 - ❖ Mitigating this risk often requires redundancy measures, backup solutions, and disaster recovery planning.
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VIRTUALIZATION – COST OF VIRTUALIZATION

❖ **Resource contention:**

- ❖ In a virtualized environment, multiple virtual machines or containers share the same physical resources, such as CPU, memory, storage, and network bandwidth.
 - ❖ If resource allocation is not properly managed, there can be contention and performance degradation when multiple workloads demand high resources simultaneously.
 - ❖ Monitoring and managing resource usage become critical to ensure fair allocation and optimal performance.
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